

Sharp low-dimensional comparisons and structural obstructions for Nyström approximation

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Abstract

For a prescribed positive spectrum, we compare the largest possible best-subset Nyström trace error with its volume-sampling mean. We determine the sharp comparison factor in dimension four, obtain an exact formula for a continuous family with two repeated eigenvalues, and characterize equality for two-point precision spectra by the existence of an equiangular tight frame. A planar selection inequality gives a sharp regularized formula in codimension two when the dimension is divisible by three; its limiting consequence forces any dimension-independent comparison constant to be at least $3/2$. We also prove a $5/3$ comparison for arbitrary positive spectra in these codimension-two dimensions, a sharper result for paired precision spectra, and a Plücker-based upper bound on the inverse energy of a best Parseval coordinate basis. All spectra in the finite-dimensional counterexamples are strictly positive. The unrestricted sharp comparison remains unresolved.

Contribution and relation to prior work

The main contribution is a set of sharp trace-resolvent comparisons, rather than a restatement of spectral-norm subset selection. The four-dimensional constant $6/(3 + \sqrt{5})$ is obtained by reducing the entire real Grassmannian orbit to two probability three-vectors; a one-parameter orbit construction and nonnegative quartic certificates then extend the bound to every four-dimensional positive spectrum. In codimension two, a weighted planar inequality controls every positive regularization and identifies an exact finite-noise extremizer, yielding the global necessary factor $3/2$.

The closest recent rank-two results of Sengupta and Pautov [4] and Nesterenko [5] concern the smallest singular value: they provide a pair with $\|Q_I^{-1}\|_2^2 \leq n$ and describe equality for that objective. The immediate squared-Frobenius consequence is $2n$. Our trace bound is $4n/3$, attained when $3 \mid n$, and the stronger regularized statement controls $\text{tr}(\tau I + Q_I Q_I^T)^{-1}$ for every $\tau > 0$. These are distinct quantitative conclusions; we do not claim a new solution of the already solved rank-two spectral-norm conjecture.

For general rank, the established squared-Frobenius guarantee is $k(n - k + 1) = k + k(n - k)$; see Avron–Boutsidis [6] and the explicit guarantee in [7, Corollary 2]. The Plücker argument below improves this bound by $k(n - k)/6$, to $k + 5k(n - k)/6$, for the best real square coordinate basis. Its mechanism is a sharp six-minor inequality summed over octahedra in the Johnson graph. The classical frame identities, determinant averages, zero-diagonal construction, and volume-selection baseline are background, not claimed contributions. The cited comparisons establish the precise distinction from the nearest inspected results; they are not a claim of exhaustive literature priority.

1. Introduction and notation

Let K be a real symmetric positive definite $n \times n$ matrix. For $I \subset [n]$ define the Nyström trace error

$$E_K(I) = \text{tr}(K - K_{:,I} K_{I,I}^{-1} K_{I,:}).$$

For a positive spectrum $\lambda = (\lambda_1, \dots, \lambda_n)$ and $1 \leq k < n$, put

$$x_k(\lambda) = \max_{U \in O(n)} \min_{|I|=k} E_{U \text{diag}(\lambda) U^T}(I), \quad y_k(\lambda) = (k+1) \frac{e_{k+1}(\lambda)}{e_k(\lambda)},$$

where e_j is the elementary symmetric polynomial. The maximum exists by compactness of $O(n)$ and continuity of the finitely many residuals. The distinction between the worst orientation and the best subset inside that orientation is essential.

The comparison of x_k and y_k is posed in [1, Problem 4.7]. Determinantal sampling and nuclear-score selection provide the surrounding approximation framework [2]. Our results concern the orbit comparison, rather than the worst erasure objective used in classical frame design [3]. We make no claim to a complete solution of the unrestricted problem or to a classification of equiangular tight frames. The statements and proofs below form a self-contained account of results obtained on 5–6 September 2026; this manuscript is version 1.0.1, prepared on 21 September 2026. Priority beyond the cited literature has not been established.

Lemma 1.1 (Precision and volume identities). *Let $L = K^{-1}$ and $J = I^c$. Then*

$$E_K(I) = \text{tr}(L_{J,J}^{-1}), \quad \sum_{|I|=k} \det(K_{I,I}) E_K(I) = (k+1) e_{k+1}(\lambda).$$

Consequently $x_k \leq y_k$. In codimension two, writing $d_i = L_{ii}$ and $S = \text{tr } L$, the errors and their mean are

$$F_{ij} = \frac{d_i + d_j}{d_i d_j - L_{ij}^2}, \quad y_{n-2} = \frac{(n-1)S}{e_2(\lambda^{-1})}.$$

An orientation attains its volume mean precisely when all its subset errors are equal.

Proof. The residual is zero in the selected rows and columns, and its complementary block is the Schur complement of $K_{I,I}$. Block inversion identifies that complement with $L_{J,J}^{-1}$. Enlarging a principal minor by one index and using the same Schur formula gives

$$\det(K_{I,I}) E_K(I) = \sum_{j \notin I} \det K_{I \cup \{j\}, I \cup \{j\}}.$$

The sum counts each principal $(k+1)$ -minor $k+1$ times. The sum of principal j -minors equals $e_j(\lambda)$ by expansion of $\det(zI + K)$. Thus the positive weights $\det(K_{I,I})/e_k(\lambda)$ have mean error y_k , proving the claims. In codimension two one may also sum $\det(L_{\{i,j\}}) F_{ij} = d_i + d_j$ and use complementary products in the elementary symmetric polynomials. $\square$

2. An exact four-dimensional orbit

Theorem 2.1. *For $A \geq B > 0$,*

$$x_2(A, A, B, B) = \frac{B \left[3(A+B) + \sqrt{(A+B)(5A-3B)} \right]}{A+3B}.$$

The corresponding mean is $6AB(A+B)/(A^2+4AB+B^2)$. The ratio y_2/x_2 increases strictly with $A/B > 1$, equals one at $A = B$, and has supremum $6/(3+\sqrt{5})$.

Lemma 2.2 (A three-coordinate minimax inequality). *For $\rho > 0$ and nonnegative three-vectors a, b with $\sum a_i = \sum b_i = 1$,*

$$\max_i (a_i + b_i + \rho \sqrt{a_i b_i}) \geq \frac{\rho + 2}{\rho + 3}.$$

Equality is attained at $a = (1 - \alpha, 0, \alpha)$, $b = (0, 1 - \alpha, \alpha)$, where $\alpha = (\rho + 3)^{-1}$.

Proof. Write the maximum as M . Summing $a_i + b_i$ gives $M \geq 2/3$; if $M \geq 1$ the claim is immediate. Otherwise set $T = 1/M \in (1, 3/2]$, $u_i = a_i/M$, $v_i = b_i/M$. The upper boundary $v = f(u)$ of $u + v + \rho \sqrt{uv} \leq 1$ on $[0, 1]^2$ is decreasing and convex. Indeed its parametrization is

$$u = (1 + z^2 + \rho z)^{-1}, \quad v = z^2(1 + z^2 + \rho z)^{-1}, \quad f'(u) = -\frac{z(2 + \rho z)}{2z + \rho},$$

and the derivative in z of the last positive fraction is $2\rho(1 + z^2 + \rho z)/(2z + \rho)^2 > 0$, while $du/dz < 0$. Moving mass between two interior coordinates with their sum fixed cannot decrease a convex sum if one chooses a suitable endpoint. Hence, under $\sum u_i = T$, the maximum of $\sum f(u_i)$ is attained at a permutation of $(1, T - 1, 0)$ and equals $1 + f(T - 1)$. Since $\sum v_i = T$, it follows that $T - 1 \leq f(T - 1)$. For $0 \leq q \leq 1$, $q \leq f(q)$ is equivalent to $(2 + \rho)q \leq 1$. Thus $T \leq 1 + 1/(2 + \rho)$, which is the required bound. The displayed vectors verify attainment. $\square$

Proof of Theorem 2.1. Use precision eigenvalues $a \geq b > 0$, and write $d = (a + b)/2$, $t = (a - b)/(a + b)$. The case $t = 0$ is immediate. Otherwise every precision matrix is $L = d(I + tH)$, with $H = 2P - I$ for a real rank-two orthogonal projection P on $\mathbb{R}^4$. Choose orthonormal columns v, w for the range of P , and set $p_{ij} = v_i w_j - v_j w_i$. Then

$$\sum_{i < j} p_{ij}^2 = 1, \quad p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0.$$

Define

$$u = (p_{12} + p_{34}, p_{13} - p_{24}, p_{14} + p_{23}), \quad v' = (p_{12} - p_{34}, p_{13} + p_{24}, p_{14} - p_{23}).$$

Both vectors are unit vectors. The three pairs of complementary principal two-blocks have H -traces $\pm 2u_j v'_j$ and common determinants $u_j^2 + (v'_j)^2 - 1$. For example, $P_{11} + P_{22} = 1 + p_{12}^2 - p_{34}^2$ and $\det P_{\{1,2\}} = p_{12}^2$ give the first pair, and the others follow by relabeling. Writing $s_j = u_j^2 + (v'_j)^2$ and $z_j = |u_j v'_j|$, the smaller error in the j th complementary pair is

$$f_j = \frac{2}{d} \frac{1 + tz_j}{1 - t^2 + t^2 s_j + 2tz_j}. \quad (2.1)$$

To select the sign, the derivative of $(1 + w)/(1 + 2w + t^2(s_j - 1))$ has sign $t^2(s_j - 1) - 1 < 0$.

Let α be the positive root of

$$t^2 \alpha^2 + (1 + 3t)\alpha - t = 0. \quad (2.2)$$

It lies in $(0, 1/3)$. Put $v_0 = (1 - t^2 \alpha)^{-1}$ and $\rho = t^{-1} + t\alpha$. Equation (2.2) gives $\alpha = (3 + \rho)^{-1}$. Lemma 2.2, applied to $u_j^2, (v'_j)^2$, produces a j with $s_j + \rho z_j \geq 1 - \alpha$. Multiplication by the positive denominators in (2.1) shows this is equivalent to $f_j \leq 2v_0/d$. Thus the upper bound covers the whole real orbit.

For attainment set $q = \sqrt{\alpha(1 - \alpha)}$ and

$$H_\alpha = \begin{pmatrix} -\alpha & 1 - \alpha & q & q \\ 1 - \alpha & -\alpha & q & q \\ q & q & \alpha & -(1 - \alpha) \\ q & q & -(1 - \alpha) & \alpha \end{pmatrix}.$$

Direct multiplication gives $H_\alpha^2 = I$ and $\text{tr } H_\alpha = 0$. Four cross-pair errors of $d(I + tH_\alpha)$ equal $2/[d(1 - t^2\alpha)]$; the smaller of its remaining pair errors is

$$\frac{2(1 + t\alpha)}{d(1 + t)(1 - t + 2t\alpha)},$$

which is the same value by (2.2). The sixth is larger. Solving (2.2) and simplifying gives

$$\frac{2v_0}{d} = \frac{3(a + b) + \sqrt{(a + b)(5a - 3b)}}{a(a + 3b)}.$$

Set $a = 1/B, b = 1/A$ to obtain the formula.

For the ratio, let $w = \sqrt{1 + 4t} \in [1, \sqrt{5})$. It equals

$$R(w) = \frac{6(3 - w)(w^2 + 3)}{47 + 2w^2 - w^4}.$$

The denominator is positive. With $N = (3 - w)(w^2 + 3)$ and $D = 47 + 2w^2 - w^4$,

$$N'D - ND' = (w - 1)p(w), \quad p(1 + h) = 32(h - 3/4)^2 + 46 + h^3(4 - h^2) > 0$$

for $0 \leq h < \sqrt{5} - 1$, as is verified by expansion. Thus R increases strictly away from $w = 1$. Its endpoint values are one and the asserted limiting constant. $\square$

3. Equality and equiangular tight frames

Theorem 3.1. *Let $n \geq 3$, $1 \leq r < n$, and $a, b > 0$ with $a \neq b$. For the covariance spectrum reciprocal to (a^{n-r}, b^r) , equality $x_{n-2} = y_{n-2}$ holds if and only if there is a real rank- r orthogonal projection satisfying*

$$P_{ii} = r/n, \quad P_{ij}^2 = \frac{r(n-r)}{n^2(n-1)} \quad (i \neq j).$$

Equivalently there is a real equiangular tight frame of n unit vectors in $\mathbb{R}^r$. The complex statement is identical with absolute squares and Hermitian projections.

Proof. Write $L = aI + (b - a)P$, $S = (n - r)a + rb$, and $c = a + b$. Then $L^2 = cL - abI$. Equality in Lemma 1.1 means every pair error is $h = (n - 1)S/e_2(L)$. Summing $h(d_i d_j - |L_{ij}|^2) = d_i + d_j$ over $j \neq i$ gives

$$[h(S - c) - (n - 2)]d_i = S - hab.$$

As $e_2(L) = [S(S - c) + nab]/2$, the coefficient on the left is $nD/[2e_2(L)]$ and the right side is $SD/[2e_2(L)]$, where

$$D = (n - r)(n - r - 1)a^2 + r(r - 1)b^2 + 2(r - 1)(n - r - 1)ab > 0.$$

Hence $d_i = S/n$ for every i . The pair equations then force all $|L_{ij}|^2$ equal, and $P^2 = P$ gives the stated values. Conversely those values make all pair errors equal. Factoring $P = V^*V$, $VV^* = I_r$, shows that the columns of $\sqrt{n/r}V$ are exactly the required unit-norm tight frame. $\square$

For $n = 4, r = 2$ a real such frame does not exist: its four rank-one projectors would have nonsingular Gram matrix $(1 - c^2)I + c^2J$, $c^2 = 1/3$, and hence be linearly independent in the three-dimensional space of real symmetric 2×2 matrices. In contrast

$$C = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & -i & i \\ 1 & i & 0 & -i \\ 1 & -i & i & 0 \end{pmatrix}, \quad C^2 = 3I,$$

makes $(I + C/\sqrt{3})/2$ a complex equality projection. Thus the real and complex orbit problems differ even in the smallest intermediate case.

4. The sharp factor for every four-dimensional spectrum

Theorem 4.1. *Every positive real spectrum of length four satisfies*

$$1 \leq \frac{y_2(\lambda)}{x_2(\lambda)} < \frac{6}{3 + \sqrt{5}}, \quad \sup_{\lambda > 0} \frac{y_2(\lambda)}{x_2(\lambda)} = \frac{6}{3 + \sqrt{5}}.$$

Proof. Put $\rho = (3 + \sqrt{5})/6$, so $\rho^2 = \rho - 1/9$ and $5/6 < \rho < 8/9$. Order the precision eigenvalues as $s \geq p \geq r \geq q > 0$, and set $\tau = s + p + r + q$, $e = sp + sr + sq + pr + pq + rq$, $y = 3\tau/e$, $h = \rho y$. For $q \leq w \leq p$ define $u = p + q - w$, $v = \sqrt{(p - w)(w - q)}$, and

$$L(w) = \frac{1}{2} \begin{pmatrix} u + r & u - r & v & v \\ u - r & u + r & v & v \\ v & v & w + s & w - s \\ v & v & w - s & w + s \end{pmatrix}.$$

In the symmetric-pair and antisymmetric-pair orthonormal basis, this has a 2×2 block of trace $p + q$ and determinant pq , together with scalar blocks r, s . It has exactly the prescribed precision spectrum. Its errors are

$$F_{12} = 1/r + 1/(p + q - w), \quad F_{34} = 1/s + 1/w, \quad F_{\text{cross}} = \frac{2\tau}{A + (r - s)w}, \quad A = pq + rs + s(p + q),$$

with four cross copies. The first and cross expressions are nondecreasing; the second is decreasing.

At $w = p$, $F_{12} \geq y$ (it is the largest pair error of a diagonal orientation), and $F_{\text{cross}} \geq y$ because

$$2sp + 2rq - (s + p)(r + q) = (s - r)(p - q) + (s - q)(p - r) \geq 0.$$

At the other endpoint, $F_{34}(q) \geq (8/9)y > h$. Indeed e/τ increases in each precision value: fixing the sum S and second symmetric polynomial E of the others, the derivative of $(zS + E)/(z + S)$ is $(S^2 - E)/(z + S)^2 > 0$. Lowering p, r to q therefore gives

$$\frac{F_{34}(q)}{y} \geq \frac{(s + q)^2}{s(s + 3q)} \geq \frac{8}{9},$$

the last inequality being $(s - 3q)^2 \geq 0$.

If $h \leq 1/s$, choose $w = p$. Otherwise put $W = s/(hs - 1) > q$. If $W > p$, again $w = p$ works; if $W = p > q$, decrease w slightly to make the decreasing error strict while preserving the other two strict margins. The case $p = q$ already follows from the endpoint inequalities. We are left with $q < W < p$, for which $F_{34}(W) = h$.

The inequality $F_{\text{cross}}(W) \geq h$ is equivalent to

$$P = h^2 s A - h[pq + s(3\tau - r)] + 2\tau \leq 0.$$

Using $h = 3\rho\tau/e$ gives $e^2 P/\tau = P_0 + \rho P_1$, where

$$P_0 = 2e^2 - \tau s A, \quad P_1 = 9\tau s A - 3e[pq + s(3\tau - r)].$$

If $h \leq 1/r$, then $F_{12}(W) > h$ already. Otherwise its weak comparison is equivalent to

$$T = (p + q)h^2 sr - h[(p + q)(s + r) + 2sr] + \tau \leq 0,$$

and $e^2 T/\tau = T_0 + \rho T_1$, with

$$T_0 = e^2 - \tau sr(p + q), \quad T_1 = 9\tau sr(p + q) - 3e[(p + q)(s + r) + 2sr].$$

In the independent gaps $a = s - p$, $b = p - r$, $c = r - q$, $d = q$, all four polynomials

$$G_0 = -6P_0 - 5P_1, \quad G_1 = -P_1, \quad J_0 = -6T_0 - 5T_1, \quad J_1 = -T_1$$

have nonnegative coefficients. Their complete expansion is Table 1; in particular the coefficients of d^4 in G_0, J_0 are 24, 12. As $d > 0$,

$$P_0 + \rho P_1 = -G_0/6 - (\rho - 5/6)G_1 < 0, \quad T_0 + \rho T_1 = -J_0/6 - (\rho - 5/6)J_1 < 0.$$

Thus both other errors at W are strictly above h , and a sufficiently small decrease of W makes all six strictly above it. This proves $x_2 > \rho y_2$. Sharpness follows from Theorem 2.1 as $A/B \rightarrow \infty$. $\square$

Table 1: Coefficients of $a^i b^j c^\ell d^t$ in the four quartic certificates. Omitted coefficients vanish.

i	j	ℓ	t	G_0	G_1	J_0	J_1
3	1	0	0	6	0	0	0
3	0	1	0	12	0	0	0
3	0	0	1	18	0	0	0
2	2	0	0	12	0	9	3
2	1	1	0	60	6	12	6
2	1	0	1	72	6	30	12
2	0	2	0	27	3	27	9
2	0	1	1	84	12	66	24
2	0	0	2	51	9	48	18
1	3	0	0	6	0	18	6
1	2	1	0	93	21	21	15
1	2	0	1	96	24	60	30
1	1	2	0	132	42	30	24
1	1	1	1	330	120	102	78
1	1	0	2	183	81	93	63
1	0	3	0	45	21	27	15
1	0	2	1	204	102	78	60
1	0	1	2	279	153	75	81
1	0	0	3	120	72	24	36
0	4	0	0	0	0	9	3
0	3	1	0	45	15	9	9
0	3	0	1	42	18	30	18
0	2	2	0	108	48	0	18
0	2	1	1	240	132	36	66
0	2	0	2	126	90	42	54
0	1	3	0	81	51	9	21
0	1	2	1	282	222	48	102
0	1	1	2	321	315	63	153
0	1	0	3	120	144	24	72
0	0	4	0	18	18	9	9
0	0	3	1	84	108	42	54
0	0	2	2	138	234	69	117
0	0	1	3	96	216	48	108
0	0	0	4	24	72	12	36

5. A sharp planar selection inequality

For a real $n \times k$ Parseval matrix Q ($Q^T Q = I_k$), define

$$h(Q) = \min_{|I|=k, \det Q_I \neq 0} \|Q_I^{-1}\|_F^2, \quad M_{n,k} = \sup_{Q^T Q = I_k} h(Q).$$

We use a supremum in the definition so that no continuity assertion at singular bases is needed.

Lemma 5.1. *Suppose $t_i \geq 0$, $\sum_i t_i = 1$, and $\sum_i t_i v_i v_i^T = I_2$ for vectors $v_i \in \mathbb{R}^2$. For every real $c \leq 3/4$ there are distinct indices i, j with*

$$\det(v_i, v_j)^2 \geq c(\|v_i\|^2 + \|v_j\|^2) + 3 - 4c.$$

Proof. Discard zero weights and zero vectors, write $v_i = \|v_i\|(\cos \theta_i, \sin \theta_i)$, and put

$$u_i = (\cos 2\theta_i, \sin 2\theta_i), \quad d_i = \|v_i\|^{-2}, \quad p_i = t_i \|v_i\|^2 / 2.$$

Then $\sum p_i = 1$, $\sum p_i u_i = 0$, $\sum p_i d_i \leq 1/2$. Among probability weights on the same directions with zero mean choose a minimizer of the d -mean with smallest support. Such a minimizer exists on a compact polytope. It has at most three points: a dependence among more than three vectors $(1, u_i)$ permits a feasible perturbation; minimality forces zero change in the objective, and movement until a coefficient vanishes contradicts minimal support. Its support has at least two points.

Write $D = 3 - 4c \geq 0$ and $\sigma_{ij} = \sin^2(\theta_i - \theta_j)$. The desired inequality is

$$\sigma_{ij} \geq c(d_i + d_j) + D d_i d_j. \quad (5.1)$$

On a two-point support the doubled directions are antipodal, with weights $1/2$. Thus $\sigma_{12} = 1$ and $S = d_1 + d_2 \leq 1$. Since $d_1 d_2 \leq S^2/4$ and the quadratic $cS + DS^2/4$ is convex on $[0, 1]$ with endpoint values $0, 3/4$, (5.1) holds.

On a three-point support the directions are affinely independent and $0 < p_i < 1/2$. For the pair i, j , let ℓ be the omitted index and put $b_{ij} = 1 - 2p_\ell > 0$. Then

$$\sum b_{ij} = 1, \quad \sum_{j \neq i} b_{ij} = 2p_i, \quad \sigma_{ij} = \frac{b_{ij}}{4p_i p_j}.$$

Furthermore

$$\sum_{i < j} b_{ij} \sigma_{ij} \geq \frac{3}{4}.$$

Indeed, after multiplication by $4p_1 p_2 p_3$, the required numerator difference is $1 - 4e_2(p) + 9e_3(p)$. Ordering $p_1 \geq p_2 \geq p_3$, it equals

$$(p_1 - p_2)^2(p_1 + p_2 - p_3) + p_3(p_1 - p_3)(p_2 - p_3) \geq 0.$$

For $h = \sum p_i d_i \leq 1/2$ one also has

$$\sum_{i < j} b_{ij} d_i d_j = h^2 - \left\| \sum_i p_i d_i u_i \right\|^2 \leq h^2.$$

If every pair failed (5.1), averaging with b_{ij} would give $3/4 < 2ch + Dh^2 \leq 3/4$. The last bound follows from convexity on $[0, 1/2]$, with endpoint values $0, 3/4$. This contradiction proves the lemma, including negative c . $\square$

Corollary 5.2. *If $Q \in \mathbb{R}^{n \times 2}$ is Parseval and has n_+ nonzero rows, then $h(Q) \leq 4n_+/3$. For every $r \geq 1$,*

$$M_{3r,2} = 4r.$$

Proof. Delete the zero rows and apply Lemma 5.1 to $v_i = \sqrt{n_+}q_i$, $t_i = 1/n_+$, $c = 3/4$. The chosen pair is nonsingular and has inverse Frobenius cost at most $4n_+/3$. For $n = 3r$, take r copies of each row $\sqrt{2/n}(\cos \theta, \sin \theta)$ for $\theta = 0, \pi/3, 2\pi/3$. This is Parseval. Every nonsingular pair has Gram eigenvalues $1/n, 3/n$, so its cost is $4n/3$. $\square$

The weighted three-row problem also has an exact answer, which records how unequal row multiplicities affect the extremum.

Proposition 5.3. *Let $B^T B = I_2$ for a real 3×2 matrix, and divide its i th row by $\sqrt{t_i}$, where $t_i > 0$, obtaining A . Put $T = \sum_i t_i$. If $c > 0$ is the unique solution of*

$$\sum_{i=1}^3 \frac{t_i}{c + 2t_i} = 1,$$

then some invertible row pair of A has cost at most $T + c \leq 4T/3$. For each fixed positive t , the bound $T + c$ is attained by a suitable B .

Proof. Write $BB^T = I - vv^T$, $z_i = v_i^2$, and $S = \sum t_i z_i > 0$. The pair omitting j is invertible exactly when $z_j > 0$, and Sherman–Morrison gives cost $F_j = T - 2t_j + S/z_j$. If all eligible costs exceeded $T + c$, then $z_j < S/(c + 2t_j)$ for all j , including $z_j = 0$, and multiplication by t_j and summation would give $S < S$. Existence and uniqueness of c follow from strict decrease of the defining sum from $3/2$ to zero. Concavity of $t/(c + 2t)$ and Jensen at $c = T/3$ give $c \leq T/3$. Finally choose z_i proportional to $(c + 2t_i)^{-1}$, and choose the columns of B as an orthonormal basis of $(\sqrt{z_i})^\perp$. The same formula then makes every $F_j = T + c$. $\square$

6. Codimension two: exact regularization and a global obstruction

Theorem 6.1. *Let $n = 3r$, $r \geq 1$, and $k = n - 2$. For $a \geq b > 0$,*

$$x_k(a^{n-2}, b^2) = \frac{nab}{a + (n-1)b} + \frac{nab}{3a + (n-3)b}.$$

Writing $t = b/a$, the ratio is

$$\frac{y_k}{x_k} = \frac{n-1}{2n} \frac{3 + (4n-6)t + (n-1)(n-3)t^2}{1 + 2(n-2)t + (n-2)(n-3)t^2/2}.$$

It is nonincreasing in $t \in (0, 1]$ and has supremum $3(n-1)/(2n)$. In particular any universal constant satisfying $y_k(\lambda) \leq Cx_k(\lambda)$ must satisfy $C \geq 3/2$.

Proof. For $a > b$ put $\delta = 1/a$, $\gamma = 1/b - 1/a$. Every precision orientation is $L = \delta I + \gamma QQ^T$, $Q^T Q = I_2$. For any $\tau > 0$, apply Lemma 5.1 to $v_i = \sqrt{n}q_i$, $t_i = 1/n$, with

$$z = n\tau, \quad c = \frac{3 - z^2}{2(z + 2)} \leq \frac{3}{4}, \quad D = 3 - 4c = \frac{z(2z + 3)}{z + 2}.$$

If $S = \|v_i\|^2 + \|v_j\|^2$, $P = \det(v_i, v_j)^2$, the lemma gives $P \geq cS + D$. Direct multiplication by positive denominators makes this equivalent to

$$\text{tr}(\tau I_2 + Q_J Q_J^T)^{-1} = \frac{n(2z + S)}{z^2 + zS + P} \leq \frac{n}{z + 1} + \frac{n}{z + 3}.$$

Use $\tau = \delta/\gamma$ and divide by γ . This gives an all-orbit upper bound $1/(\delta + \gamma/n) + 1/(\delta + 3\gamma/n)$. The three-direction repeated frame in Corollary 5.2 attains it: different-direction pairs have these two Gram eigenvalues, while same-direction pairs have eigenvalues 0, $4/n$ and a larger inverse trace, since

$$\frac{1}{\delta} + \frac{1}{\delta + 4d} - \frac{1}{\delta + d} - \frac{1}{\delta + 3d} = \frac{d}{\delta(\delta + d)} - \frac{d}{(\delta + 3d)(\delta + 4d)} \geq 0.$$

Here $d = \gamma/n$. The case $a = b$ yields $2a$ on both sides.

Elementary symmetric expansion gives

$$y_k = \frac{(n-1)ab[2a + (n-2)b]}{a^2 + 2(n-2)ab + (n-2)(n-3)b^2/2}.$$

Division gives the stated ratio. For its numerator N and denominator D_n without the constant factor, $N'D_n - ND'_n = (n-3)(t-1)((n-2)t+2) \leq 0$. The limit at zero is $3(n-1)/(2n)$. Taking the supremum over $n = 3r$ gives $3/2$, using only strictly positive spectra at every finite stage. $\square$

Theorem 6.2 (Unequal tails). *Let $n = 3r$, $k = n - 2$, and $\lambda_1 \geq \dots \geq \lambda_n > 0$. Put $a = \lambda_{n-2}$, $s = \lambda_{n-1} + \lambda_n$, $H = 2ns/3$. Then*

$$x_k(\lambda) \geq \frac{aH}{a + H - s}, \quad \frac{y_k(\lambda)}{x_k(\lambda)} \leq \frac{3(n-1)}{2n} \left[1 + \left(\frac{2n}{3} - 1 \right) \frac{s}{a} \right].$$

For every fixed positive top spectrum and fixed positive β_1, β_2 ,

$$\limsup_{\varepsilon \downarrow 0} \frac{y_k(a_1, \dots, a_{n-2}, \varepsilon\beta_1, \varepsilon\beta_2)}{x_k(a_1, \dots, a_{n-2}, \varepsilon\beta_1, \varepsilon\beta_2)} \leq \frac{3(n-1)}{2n}.$$

This boundary constant is sharp.

Proof. We first recall an elementary zero-diagonal construction. A nonzero traceless real symmetric matrix has positive and negative eigenvalues. A unit vector in their two-dimensional eigenspace span can be chosen to have zero quadratic form. Its orthogonal-complement compression remains traceless, so induction produces an orthonormal basis with zero diagonal. Applying this to $\text{diag}(-s, \lambda_{n-1}, \lambda_n)$ yields an orthogonal matrix $[q \ R_0]$ such that, for $B = \text{diag}(\lambda_{n-1}, \lambda_n)$,

$$(R_0 B R_0^T)_{ii} = s q_i^2.$$

Every q_i is nonzero, since otherwise the corresponding R_0 row would be unit and have a positive quadratic form. If row ℓ is omitted, Sherman–Morrison gives

$$\text{tr} [B((R_0)_J^T (R_0)_J)^{-1}] = s + \frac{r_\ell^T B r_\ell}{q_\ell^2} = 2s.$$

Replace each row by r copies divided by $\sqrt{r}$, giving $R^T R = I_2$, and complete to $[W \ R] \in O(n)$. Set $K = W \text{diag}(\lambda_1, \dots, \lambda_{n-2}) W^T + R B R^T$. Then

$$K^{-1} \leq a^{-1}(I - R R^T) + R B^{-1} R^T =: A_*.$$

Principal restriction preserves this order, and positive definite inversion reverses it.

If a complementary pair uses two distinct row types, put $G = R_J B^{-1} R_J^T$, $X = G^{-1}$, $T = I - R_J R_J^T \geq 0$. The preceding identities give $\text{tr} X = H$ and $\text{tr}(X R_J R_J^T) = s$. Hence $X^{1/2} T X^{1/2} \leq (H - s)I$, so $(A_*)_J \leq [1 + (H - s)/a]G$. Its inverse trace is at least $aH/(a + H - s)$. If the two types

coincide, $(A_*)_J = a^{-1}I + R_J(B^{-1} - a^{-1}I)R_J^T$ has an eigenvalue $1/a$; the other inverse eigenvalue is at least λ_n . Its trace is at least $a + \lambda_n$, which is at least $aH/(a + H - s)$ because

$$(a + \lambda_n)(a + H - s) - aH = a(a - \lambda_{n-1}) + \lambda_n(H - s) \geq 0.$$

This controls every pair. Finally $e_{k+1}(\lambda) \leq se_k(\lambda)$ because every $(k + 1)$ -subset meets the two tail coordinates; counting products proves the inequality. This yields the comparison. For fixed coefficients and tails scaled by ε , $y_k/\varepsilon \rightarrow (n - 1)(\beta_1 + \beta_2)$ and the displayed lower bound on x_k/ε tends to $2n(\beta_1 + \beta_2)/3$. Sharpness follows from Theorem 6.1 with equal top and tail values. $\square$

7. Paired precision spectra on a ternary group

Theorem 7.1. *Let $n = 3^d$, $d \geq 1$, $p = (n - 1)/2$, and $k = n - 2$. Suppose the precision spectrum is*

$$(\delta + t, \delta + w_1, \delta + w_1, \dots, \delta + w_p, \delta + w_p), \quad \delta > 0, \quad t, w_j \geq 0.$$

Put $W = \sum_j w_j$, $w_ = \max_j w_j$, $S = n\delta + t + 2W$, $E = e_2(\lambda^{-1})$, and $z = \max(0, t - W, 3w_* - 2W - t)$. Then*

$$x_k \geq \frac{2nS}{S^2 - z^2} \geq \frac{2nS}{3E}, \quad \sup \frac{y_k}{x_k} = \frac{3(n - 1)}{2n},$$

where the supremum is over this paired class. It covers every positive precision spectrum consisting of one singleton and equal pairs.

Proof. Use the group $G = (\mathbb{Z}/3\mathbb{Z})^d$ and one representative a of each nonzero pair $\{a, -a\}$. The constant column $u = n^{-1/2}\mathbf{1}$ and columns

$$c_a(x) = \sqrt{2/n} \cos(2\pi a \cdot x/3), \quad s_a(x) = \sqrt{2/n} \sin(2\pi a \cdot x/3)$$

form an orthonormal real basis. To verify this, write trigonometric functions as exponentials and use $\sum_x \exp(2\pi i b \cdot x/3) = n$ for $b = 0$, and zero otherwise; the latter sum factors over coordinates. Define

$$L = \delta I + tuu^T + \sum_a w_a (c_a c_a^T + s_a s_a^T).$$

It has the required spectrum and diagonal S/n . For $x \neq y$,

$$nL_{xy} = t + 2W - 3 \sum_{a: a \cdot (x-y) \neq 0} w_a.$$

Its absolute value is at least z . The case $z = t - W > 0$ follows by bounding the subset sum by W . In the case $z = 3w_* - 2W - t > 0$, write $T = W - w_*$. If the largest weight is outside the sum the expression is at least $t + 2w_* - T \geq z$; if inside, it is at most $-z$. The alternatives cannot both be positive. Thus every pair has inverse trace at least $2nS/(S^2 - z^2)$, whose denominator is positive since $z \leq \max(t, W) < S$.

Write $Q = \sum_i \lambda_i^{-2}$. The second comparison is equivalent to $z^2 \geq (3Q - S^2)/2$. Expansion yields

$$\frac{3Q - S^2}{2} = t^2 - 2tW + 3 \sum w_a^2 - 2W^2 + (3 - n)\delta(t + 2W) + \frac{3n - n^2}{2}\delta^2.$$

The last two terms are nonpositive. If $z = 0$, then $t \leq W$ and $3w_* \leq t + 2W$, giving $3 \sum w_a^2 \leq (t + 2W)W$ and a remaining upper bound $t(t - W) \leq 0$. If $z = t - W > 0$, subtracting the first four terms from z^2 gives $3(W^2 - \sum w_a^2) \geq 0$. Otherwise $z = w_* - 2T - t$, and, with $R = \sum_{a \neq a_*} w_a^2 \leq T^2$, the difference is $6T^2 + 6tT - 3R \geq 0$. This proves the inequality.

Sharpness follows either from Theorem 6.1 or directly by taking $t = 0, w_1 = 1, w_j = 0$ for $j > 1$ and $\delta \downarrow 0$: Corollary 5.2 gives $x_k \leq 4n/3$, the proved lower bound tends to $4n/3$, and $y_k \rightarrow 2(n - 1)$. $\square$

8. A uniform 5/3 bound in codimension two

Theorem 8.1. *For every $n = 3\ell$, $\ell \geq 1$, $k = n - 2$, and every positive real spectrum,*

$$x_k(\lambda) \leq y_k(\lambda) \leq \frac{5}{3}x_k(\lambda).$$

No multiplicity restriction is imposed. The constant is not asserted sharp.

Proof. Order and rescale the precision eigenvalues so that $\mu_1 = t \geq \mu_2 = 1$. Put $u = \sum_{i \geq 3} \mu_i > 0$, $S = t + 1 + u$, $E = e_2(\mu)$. Each tail value is at most one. The zero-diagonal argument in Theorem 6.2, applied after subtracting the mean, produces a constant-diagonal orientation of any real spectrum. Applying it to the whole precision matrix gives pair traces $2S/n$ and inverse traces at least $2n/S$. Isolating t as a one-dimensional block and making the remaining block constant-diagonal gives minimum inverse trace at least $1/t + (n - 1)/(1 + u)$. Therefore

$$\frac{y_k}{x_k} \leq \frac{S^2}{2E}, \quad \frac{y_k}{x_k} \leq \frac{S(1+u)}{E}. \quad (8.1)$$

For the second bound, a pair wholly in the remaining block has inverse trace at least $2(n - 1)/(1 + u)$, which is no smaller since $t \geq (1 + u)/(n - 1)$. We shall also use

$$E \geq E_0 = t(1 + u) + u, \quad E \geq E_1 = (1 + u)(t + u/2).$$

The first retains products involving the largest two values; the second follows from $\sum_{i \geq 3} \mu_i^2 \leq u$.

Tail pairing. An odd list of nonnegative numbers of total R can be paired, with one singleton, so that rounding each pair up to two copies of its maximum increases the sum by at most $R/3$. For a one-element list this is immediate. For an ordered longer list $c_1 \geq \dots \geq c_q$, if $c_1 \leq 2c_2$, pair the first two and induct on the rest: the increase is at most $c_1 - c_2 + (R - c_1 - c_2)/3 \leq R/3$. If $c_1 > 2c_2$, leave it alone and pair successive remaining entries; the alternating increase is at most $c_2 < R/3$. For an even list first leave one value alone and apply the odd construction, leaving two singletons. Applied to the tail, this yields total $B \leq 4u/3$ and one or two singletons according to parity. Break $(\ell - N_{\text{sing}})/2$ pairs into singleton labels. The result has ℓ singletons and $\ell - 1$ equal pairs. The quantity broken is a nonnegative integer by parity, including $\ell = 1, 2$. Let ν denote the rounded full precision spectrum. Since ν dominates μ coordinatewise, positive definite inverse order gives

$$x_k(\mu^{-1}) \geq x_k(\nu^{-1}). \quad (8.2)$$

This follows for every common orthogonal orientation before taking pair minima and orbit maxima.

A three-block construction. Suppose the largest two precision values are $a \geq b > 0$, and the positive tail, with ℓ singletons and $\ell - 1$ pairs, has total $0 < B \leq b$. There is an orientation with every pair inverse trace at least

$$n \min(h_{12}, h_{23}), \quad h_{12} = \frac{3a + 5b + 4B}{6ab + 3B(a + 3b)}, \quad h_{23} = \frac{1}{3a} + \frac{1}{b + 2B}. \quad (8.3)$$

To construct it, index coordinates by $(v, s) \in \{0, 1, 2\} \times \mathbb{Z}/\ell\mathbb{Z}$ and take

$$c_j(v, s) = \sqrt{2/n} \cos(2\pi v/3 + 2\pi j s/\ell), \quad s_j(v, s) = \sqrt{2/n} \sin(2\pi v/3 + 2\pi j s/\ell), \quad 0 \leq j < \ell.$$

For singleton tail values γ_h , choose an orthogonal $\ell \times \ell$ matrix U making $U \text{diag}(\gamma) U^T$ constant-diagonal, and put $t_h(v, s) = U_{s,h}/\sqrt{3}$. These columns form a real orthonormal basis: finite Fourier

sums in v, s annihilate every nonzero frequency, the sum frequencies of two nontrivial characters have first coordinate two, and only equal difference frequencies survive. The t_h are orthogonal to the character columns by summing in v . Assign b to c_0 , a to s_0 , each paired tail value to a character pair with $j > 0$, and the singletons to the t_h . The tail contribution is positive semidefinite with diagonal B/n , so its off-diagonal absolute values are at most B/n .

After multiplication by n , the top-two diagonal contributions at the three types are $d_1 = 2b$, $d_2 = d_3 = (3a + b)/2$. The type-12 and type-13 off-diagonal contributions are $-b$; the type-23 contribution is $-(3a - b)/2$. Their magnitudes are at least $b \geq B$. A two-block inverse trace increases with the squared off-diagonal entry at fixed diagonal, so the worst allowed tail off-diagonal is $+B$. Inversion gives the two expressions in (8.3); the type-12 determinant coefficient is $3ab + B(3a + 9b)/2$, while the type-23 eigenvalue coefficients are $3a, b + 2B$. For a same-type pair the off-diagonal coefficient is at least $d_i - B > 0$, giving inverse trace at least $n(1/(2B) + 1/(2d_i))$. This is at least nh_{23} , since

$$\frac{1}{2B} + \frac{1}{3a + b} - h_{23} = \frac{b}{2B(b + 2B)} - \frac{b}{3a(3a + b)} \geq 0.$$

Indeed the first denominator is at most $6b^2$ and the second at least $12b^2$. This proves (8.3). Both h -functions decrease with B ; the derivative numerator for h_{12} is $-9a^2 - 18ab - 45b^2$, and that for h_{23} is manifestly negative.

The small-tail regime. For $0 < u \leq 2/5$, $1 \leq t \leq 3/2$, use $a = t, b = 1, B \leq 4u/3 < 1$ in (8.3) and (8.2). The inequalities $S/[E_0 h_{12}(t, 1, 4u/3)] \leq 5/3$ and $S/[E_0 h_{23}(t, 1, 4u/3)] \leq 5/3$ reduce, by positive multiplication, to

$$\begin{aligned} F_{12} &= -9(1 - u)t^2 + (21 + 2u + 44u^2)t - 33u - 28u^2 \geq 0, \\ F_{23} &= (18 - 27u)t^2 + (-12 + u - 32u^2)t + 15u + 40u^2 \geq 0. \end{aligned}$$

The first is concave in t ; at $t = 1, 3/2$ it equals respectively $12 - 22u + 16u^2 \geq 16/5$ and $45/4 - 39u/4 + 38u^2 \geq 147/20$. For the second put $t = 1 + s$, $0 \leq s \leq 1/2$:

$$F_{23} = (6 - 11u + 8u^2) + s(24 - 53u - 32u^2) + s^2(18 - 27u).$$

The three coefficients are respectively at least $72/25, -58/25$, and zero, so $F_{23} \geq 43/25$. Using $y_k = (n - 1)S/E$ completes this rectangle. If instead $t \geq 3/2$, the second bound in (8.1) suffices: $S(1 + u)/E_0 \leq 5/3$ is equivalent to

$$t \geq I_0(u) := \frac{3 + u + 3u^2}{2(1 + u)},$$

and $I_0(u) \leq 3/2$ for $0 \leq u \leq 2/3$.

The remaining regimes. For $2/5 \leq u \leq 1$, use the same second bound if $t \geq I_0(u)$. For $1 \leq t \leq I_0(u)$, the first bound suffices if $G_0 = 10E_0 - 3S^2 \geq 0$. This is concave in t , equals $-2 + 8u - 3u^2 > 0$ at $t = 1$, and at I_0 equals $3S(1 + u - I_0) \geq 0$. The latter follows from $u^2 - 3u + 1 \leq 0$ on this interval. For $u \geq 1$, the second bound with E_1 works for $t \geq I_1(u) = 3/2 + u/4$. Otherwise the first works because

$$G_1 = 10E_1 - 3S^2 = -3t^2 + 4(1 + u)t + (1 + u)(2u - 3)$$

is concave and has endpoint values $2u^2 + 3u - 2 > 0$ and $15(3u^2 + 4u - 4)/16 > 0$. These regimes cover every $t \geq 1, u > 0$ and prove the theorem. $\square$

9. A Plücker bound for general Parseval bases

Theorem 9.1. For $k, m \geq 2$ and $n = k + m$,

$$M_{n,k} \leq k + \frac{5}{6}km.$$

Moreover, for $0 < \varepsilon \leq 1$,

$$x_k(1^k, \varepsilon^m) \leq \varepsilon m(1 + 5k/6).$$

Proof. Put $p_I = \det Q_I$ and $w_I = p_I^2$. Cauchy–Binet gives $\sum_I w_I = 1$. Let A be the adjacency matrix of the Johnson graph on k -subsets. For $w_I > 0$, write $QQ_I^{-1} = [I_k; C]$ after reordering rows. Parseval and Cramer’s rule yield

$$\|Q_I^{-1}\|_F^2 = k + \|C\|_F^2 = k + \frac{(Aw)_I}{w_I}.$$

Averaging with weights $w_I^2 / \sum w_J^2$ therefore gives

$$h(Q) \leq k + \frac{w^T Aw}{w^T w}. \quad (9.1)$$

We require the elementary simplex inequality

$$2 \sum a_i^2 + 2 \sum b_i^2 + 9 \sum a_i b_i \geq 3 \quad (9.2)$$

for nonnegative three-vectors of sum one. If two coordinates overlap positively, move their entries by $(a_i, a_j, b_i, b_j) \mapsto (a_i + t, a_j - t, b_i - t, b_j + t)$. The expression is strictly concave in t with quadratic coefficient -10 , so an endpoint does not increase it and removes an overlap. Iterate. If the supports are disjoint with sizes r, s , then $r + s \leq 3$ and the expression is at least $2/r + 2/s \geq 3$. If exactly one coordinate overlaps and one support is a singleton, writing the other mass there as t gives lower bound $3 + 7t + 3t^2$. Otherwise both supports have size two: $a = (x, 1 - x, 0)$, $b = (y, 0, 1 - y)$, and the expression equals $3 + 4(x + y - 1/2)^2 + xy \geq 3$.

For six real numbers satisfying $p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0$, set $w_{ij} = p_{ij}^2$ and normalize $\sum p_{ij}^2 = 1$, excluding the trivial zero case. Define the two unit three-vectors u, v as in Theorem 2.1, and set $a_i = u_i^2, b_i = v_i^2$. If

$$N = \sum p_{ij}^4, \quad B = p_{12}^2 p_{34}^2 + p_{13}^2 p_{24}^2 + p_{14}^2 p_{23}^2,$$

then expansion gives $N = \frac{1}{8} \sum (a_i^2 + 6a_i b_i + b_i^2)$ and $B = \frac{1}{16} \sum (a_i - b_i)^2$. Thus $13N + 6B \geq 3$ by (9.2). On the octahedral adjacency graph,

$$w^T A_{\text{oct}} w = 1 - N - 2B \leq \frac{10}{3}N.$$

Homogeneity removes the normalization.

Choose a $(k - 2)$ -set R and a disjoint four-set D . The six minors indexed by R together with two elements of D satisfy the preceding Plücker relation, with the standard alternating signs. If R is dependent they all vanish; otherwise expanding along a column basis adapted to R reduces them to two-dimensional minors, proving the relation directly. Each Johnson vertex occurs in $\binom{k}{2} \binom{m}{2}$ such octahedra, while each edge occurs in $(k - 1)(m - 1)$. Summing the local inequality gives

$$\frac{w^T Aw}{w^T w} \leq \frac{5}{6}km.$$

Together with (9.1), this proves the frame bound, including zero minors.

For the finite-noise statement, every matrix of the specified spectrum is $K = \varepsilon I + (1 - \varepsilon)P$, $P = QQ^T$. Since $K^2 = (1 + \varepsilon)K - \varepsilon I$, its error is

$$\frac{E_K(I)}{\varepsilon} = m - k + \text{tr}(K_{I,I}^{-1}).$$

Choose a nonsingular coordinate basis attaining $h(Q)$, and let $t_j \geq 1$ be the inverse eigenvalues of $P_{I,I}$. Then

$$\text{tr}(K_{I,I}^{-1}) = \sum_j \frac{t_j}{1 - \varepsilon + \varepsilon t_j} \leq \sum_j t_j = h(Q).$$

Minimizing and maximizing as in the definition of x_k proves the result. $\square$

Example 9.2 (A positive-spectrum separation from the four-dimensional factor). For $n = 8$, $k = 4$, $\lambda = (1, 1, 1, 1, 10^{-4}, 10^{-4}, 10^{-4}, 10^{-4})$, exact arithmetic gives

$$y_4 = \frac{20012001200020}{10016003600160001}, \quad x_4 \leq \frac{13}{7500},$$

where the notation means four copies of 10^{-4} . Consequently

$$\frac{y_4}{x_4} \geq \frac{150090009000150000}{130208046802080013} > \frac{23}{20} > \frac{6}{3 + \sqrt{5}}.$$

The first strict difference is $7015103555159701/2604160936041600260 > 0$. For the second, the polynomial $z^2 - 9z + 9$, whose smaller root is $6/(3 + \sqrt{5})$, has value $-11/400$ at $23/20$. This example shows that the sharp four-dimensional factor cannot hold unchanged in every dimension. The $3/2$ obstruction of Theorem 6.1 is stronger asymptotically.

10. Scope and open questions

The sharp four-dimensional comparison, exact codimension-two families, and the $3/2$ necessary global factor concern different quantifiers. None implies that $3/2$ is a universal upper bound. Even in dimensions divisible by three, the unrestricted codimension-two factor remains between its exhibited lower obstruction and $5/3$. The paired-spectrum theorem does not remove all multiplicity restrictions, and the Plücker bound remains quadratic in balanced dimensions. Establishing a sharp all-spectrum comparison or deciding whether a dimension-independent comparison exists requires additional arguments.

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